

Mathematical Formulas & Question Types

Spatial & Gray-Level Resolution

1. Spatial Resolution

Represents the level of detail (fineness of image representation).

Formulas:

$$\Delta x = \text{Image Width (W)} / \text{Number of Columns (M)}$$

$$\Delta y = \text{Image Height (H)} / \text{Number of Rows (N)}$$

$$P = M \times N \quad (\text{Total number of pixels})$$

$$R_s = \Delta x \times \Delta y \quad (\text{Area per pixel})$$

Question Types:

1. Find pixel size given image size and pixel count.
2. Find number of pixels in a given resolution.
3. Compare two digitizations and decide which has better resolution.

2. Gray-Level Resolution

Represents the number of intensity levels that each pixel can take.

Formulas:

$$L = 2^k \quad (\text{Number of gray levels, } k = \text{bits per pixel})$$

$$\text{Range} = 0 \text{ to } (L-1)$$

$$\Delta I = (\text{Max Intensity} - \text{Min Intensity}) / L \quad (\text{Quantization step size})$$

$$k = \log_2(L) \quad (\text{Bits required for given gray levels})$$

Question Types:

1. Find number of gray levels for given bits.
2. Find bits required for a given number of gray levels.

3. Find quantization step size for given intensity range & bits.

3. Storage Requirement

Formulas:

$$B = (M \times N) \times k \quad (\text{Total bits required})$$

$$S = B / 8 \quad (\text{Total bytes required})$$

$$1 \text{ KB} = 1024 \text{ bytes}, 1 \text{ MB} = 1024 \text{ KB}$$

Question Types:

1. Compute storage size of an image with given resolution & bits.
2. Find how many images can fit in given memory.
3. Compare memory needed for two images.

4. Combined Questions

Example Patterns:

1. Image digitization: Find pixel size, gray levels, and storage for given dimensions & bits.
2. Resolution change effect: Compare pixels when resolution increases.
3. Bit-depth change effect: Calculate % reduction in storage.

$$\text{Reduction \%} = ((\text{Old bits} - \text{New bits}) / \text{Old bits}) \times 100$$

5. Extra Useful Relations

$$L = 2^k \quad (\text{Number of gray levels per pixel})$$

$$\text{DR} = 20 \log_{10}(L-1) \quad (\text{Dynamic range in dB})$$

$$\text{CR} = \text{Uncompressed Size} / \text{Compressed Size} \quad (\text{Compression ratio})$$

Bit-plane representation: Each image can be decomposed into k binary images (bit-planes).

Mathematical Formulas: Zooming & Shrinking in Images

1. Image Scaling Basics

Scaling (zoom/shrink) is done by multiplying coordinates by a scaling factor.

$$(x', y') = (sx * x, sy * y)$$

If $sx, sy > 1 \rightarrow$ Zooming (magnification)

If $0 < sx, sy < 1 \rightarrow$ Shrinking (reduction)

2. Zooming (Enlargement)

New image size: $M' = M * sy$, $N' = N * sx$

Nearest Neighbor (Pixel Replication): Each pixel is copied $sx \times sy$ times.

Bilinear Interpolation Formula:

$$f(x,y) \approx (1-\alpha)(1-\beta)f(i,j) + \alpha(1-\beta)f(i+1,j) + (1-\alpha)\beta f(i,j+1) + \alpha\beta f(i+1,j+1)$$

where $\alpha = x - i$, $\beta = y - j$

3. Shrinking (Reduction)

New image size: $M' = M / sy$, $N' = N / sx$

Nearest Neighbor (Subsampling): Take every sx -th and sy -th pixel.

Averaging Method:

$$f'(i,j) = (1 / (sx * sy)) \sum \sum f(sy*i + m, sx*j + n)$$

where summation $m=0..sx-1$, $n=0..sy-1$

4. General Scaling Matrix (Geometric Transformation)

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} sx & 0 & 0 \\ 0 & sy & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

5. Storage & Resolution Effect

Zooming: Increases image size (more pixels) but may create redundancy.

Shrinking: Decreases image size (fewer pixels), may cause information loss.

6. Exam Question Patterns

1. Compute new size of an image after zooming/shrinking.
2. Apply nearest neighbor / bilinear interpolation formula for a given pixel.
3. Find new storage requirement after scaling.
4. Compare quality between replication, bilinear, and averaging methods.

Mathematical Formulas: Relationship Between Pixels

1. Basic Definitions

A pixel at location (x, y) is denoted by: $p(x, y)$.

The neighborhood of a pixel depends on connectivity type.

2. Types of Neighborhood

1. 4-neighbors (N4): $N4(p) = \{(x+1, y), (x-1, y), (x, y+1), (x, y-1)\}$
2. Diagonal neighbors (ND): $ND(p) = \{(x+1, y+1), (x+1, y-1), (x-1, y+1), (x-1, y-1)\}$
3. 8-neighbors (N8): $N8(p) = N4(p) \cup ND(p)$

3. Connectivity

Defines whether pixels are connected (part of same region).

- 4-connectivity: pixels are connected if they share an edge.
- 8-connectivity: pixels are connected if they share edge or corner.
- m-connectivity: combination of 4 and diagonal connections, avoids ambiguity.

4. Adjacency

Two pixels p and q are adjacent if:

- They are neighbors (depending on 4, 8, or m-connectivity).
- Their gray-level values satisfy a condition (e.g., both belong to same set V).

Mathematically:

$\text{Adj}(p,q) = 1$, if $q \in N(p)$ and $f(p), f(q) \in V$; else 0

5. Path & Connectivity

A path from p to q : $p = p_0, p_1, p_2, \dots, p_n = q$ such that p_i is adjacent to p_{i-1} .

If such a path exists $\rightarrow p$ and q are connected.

6. Region & Boundary

Region (R): a set of pixels with similar properties.

Boundary (∂R): pixels in region R that have at least one neighbor outside R .

Mathematically: $\partial R = \{ p \in R \mid N(p) \not\subset R \}$

7. Distance Measures

1. Euclidean Distance (ED): $DE(p,q) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$

2. City-block Distance (D_4): $D_4(p,q) = |x_1 - x_2| + |y_1 - y_2|$

3. Chessboard Distance (D_8): $D_8(p,q) = \max(|x_1 - x_2|, |y_1 - y_2|)$

8. Exam Question Patterns

1. Write formulas for 4, 8, diagonal neighbors.
2. Define adjacency & connectivity mathematically.
3. Find distance between two pixels using Euclidean, City-block, and Chessboard metrics.
4. Define region and boundary mathematically.
5. Given pixel coordinates, compute neighborhood and distance.